NINE LOCAL PROPERTIES OF MORPHISMS OF ALGEBRAIC SPACES ARE INSENSITIVE TO A THICKENING OF THE TARGET

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2155-2171 including the nine enumerated items; proof span is lines 2173-2247.

Editorial note. The statement ends with the author's own placeholder item 'add more here', which is retained verbatim because it belongs to the statement environment being collected.

Editorial note. The proof cites the schemes version of the lemma plus several morphism-property results, all supplied as non-load-bearing navigation stubs.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Morphisms of thickenings of algebraic spaces, surjective etale charts, etale localness of properties of morphisms, sheaves of differentials, and the definition of properness. The chart reduction and the case analysis are retained.

Source reference. Original lemma algebra-lemma-ideal-is-squared-union-connected: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2247–2247. Referenced by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-differential-product: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1227–1240. Referenced by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-differentials-relative-immersion-section: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1180–1191. Referenced by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-thicken-property-morphisms: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2001–2015. Referenced by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma more-morphisms-lemma-properties-that-extend-over-thickenings: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2247–2247. Referenced by the retained statement or proof but not part

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of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-differentials-diagonal: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2247–2247. Referenced by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-lemma-base-change-immersions: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2247–2247. Referenced by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-limits-lemma-check-closed-infinitesimally: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2247–2247. Referenced by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-morphisms-lemma-base-change-integral: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2247–2247. Referenced by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-morphisms-lemma-base-change-monomorphism: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2247–2247. Referenced by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-morphisms-lemma-base-change-proper: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2247–2247. Referenced by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-morphisms-lemma-diagonal-morphism-finite-type: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2247–2247. Referenced by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-morphisms-lemma-finite-integral: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2247–2247. Referenced by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-morphisms-remark-immersion: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2247–2247. Referenced by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $(f, f') : (X \subset X') \to (Y \to Y')$ be a morphism of thickenings of algebraic spaces over S. Assume f and f' are locally of finite type and $X = Y \times_{Y'} X'$. Then*

(1) *f is locally quasi-finite if and only if f' is locally quasi-finite,*
(2) *f is finite if and only if f' is finite,*

(3) *f is a closed immersion if and only if f' is a closed immersion,*
(4) *$\Omega_{X/Y} = 0$ if and only if $\Omega_{X'/Y'} = 0$,*
(5) *f is unramified if and only if f' is unramified,*
(6) *f is a monomorphism if and only if f' is a monomorphism,*
(7) *f is an immersion if and only if f' is an immersion,*
(8) *f is proper if and only if f' is proper, and*
(9) *add more here.*

Proof. Choose a scheme V' and a surjective étale morphism $V' \to Y'$. Choose a scheme U' and a surjective étale morphism $U' \to X' \times_{Y'} V'$. Set $V = Y \times_{Y'} V'$ and $U = X \times_{X'} U'$. Then for étale local properties of morphisms we can reduce to the morphism of thickenings of schemes $(U \subset U') \to (V \subset V')$ and apply More on Morphisms, Lemma more-morphisms-lemma-properties-that-extend-over-thickenings. This proves (1), (4), and (5).

The properties in (2), (3), (6), (7), and (8) are stable under base change, hence if f' has property $\mathcal{P}$, then so does f. See Spaces, Lemma spaces-lemma-base-change-immersions, and Morphisms of Spaces, Lemmas spaces-morphisms-lemma-base-change-proper, spaces-morphisms-lemma-base-change-integral, and spaces-morphisms-lemma-base-change-monomorphism. Hence in each case we need only to prove that if f has the desired property, so does f'.

Case (2) follows from case (5) of Lemma lemma-thicken-property-morphisms and the fact that the finite morphisms are precisely the integral morphisms that are locally of finite type (Morphisms of Spaces, Lemma spaces-morphisms-lemma-finite-integral).

Case (3). This follows immediately from Limits of Spaces, Lemma spaces-limits-lemma-check-closed-infinitesimally.

Proof of (6). As f is a monomorphism we have $X = X \times_Y X$. We may apply the results proved so far to the morphism of thickenings $(X \subset X') \to (X \times_Y X \subset X' \times_{Y'} X')$. We conclude $\Delta_{X'/Y'} : X' \to X' \times_{Y'} X'$ is a closed immersion by (3). In fact $\Delta_{X'/Y'}$ induces a bijection $|X'| \to |X' \times_{Y'} X'|$, hence $\Delta_{X'/Y'}$ is a thickening. On the other hand $\Delta_{X'/Y'}$ is locally of finite presentation by Morphisms of Spaces, Lemma spaces-morphisms-lemma-diagonal-morphism-finite-type. In other words, $\Delta_{X'/Y'}(X')$ is cut out by a quasi-coherent sheaf of ideals $\mathcal{J} \subset \mathcal{O}_{X' \times_{Y'} X'}$ of finite type. Since $\Omega_{X'/Y'} = 0$ by (5) we see that the conormal sheaf of $X' \to X' \times_{Y'} X'$ is zero. (The conormal sheaf of the closed immersion $\Delta_{X'/Y'}$ is equal to $\Omega_{X'/Y'}$; this is the analogue of Morphisms, Lemma morphisms-lemma-differentials-diagonal for algebraic spaces and follows either by étale localization or by combining Lemmas lemma-differentials-relative-immersion-section and lemma-differential-product; some details omitted.) In other words, $\mathcal{J}/\mathcal{J}^2 = 0$. This implies $\Delta_{X'/Y'}$ is an isomorphism, for example by Algebra, Lemma algebra-lemma-ideal-is-squared-union-connected.

Proof of (7). If $f : X \to Y$ is an immersion, then it factors as $X \to V \to Y$ where $V \to Y$ is an open subspace and $X \to V$ is a closed immersion, see Morphisms of Spaces, Remark spaces-morphisms-remark-immersion. Let $V' \subset Y'$ be the open subspace whose underlying topological space $|V'|$ is the same as $|V| \subset |Y| = |Y'|$. Then $X' \to Y'$ factors through V' and we conclude that $X' \to V'$ is a closed immersion by part (3).

Case (8) follows from Lemma lemma-thicken-property-morphisms and the definition of proper morphisms as being the quasi-compact, universally closed, and separated morphisms that are locally of finite type. □